



Analyse

01 Matrix profile show the process · judge the profile against its surrogate · then threshold

Search spans, runs, families

Ctrl K

3 need you

M4 held out

< full chain

mp_span_CH4 run 14 · saved

Signal · CH4_A2 · 6.0-12.0 h · 6 h

null IAAFT 200×

≈ 4 min · local

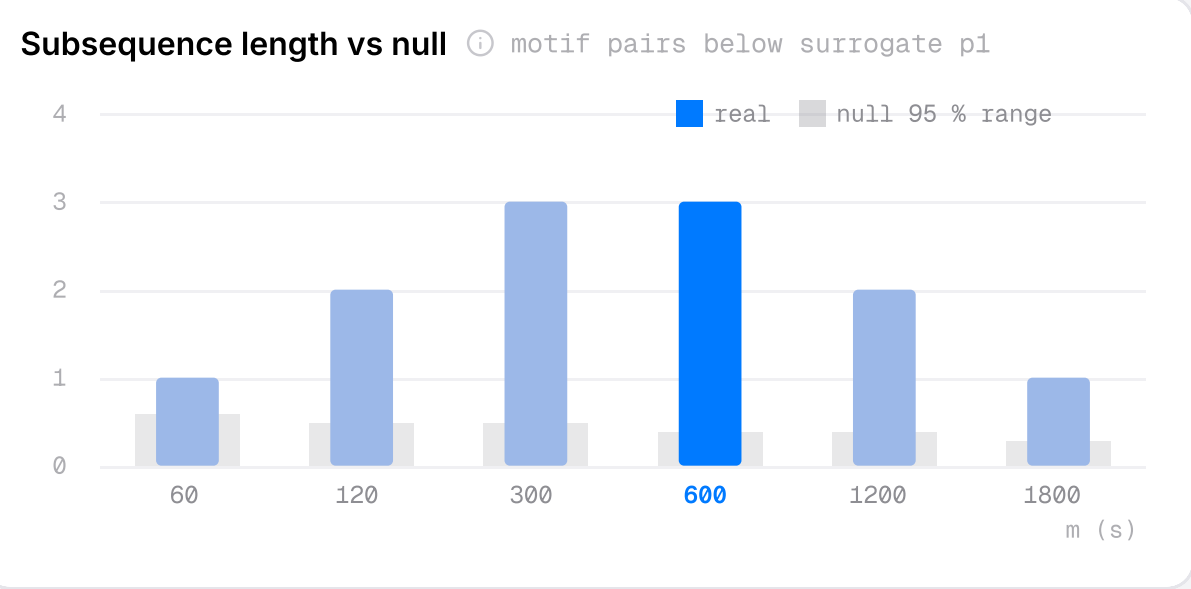
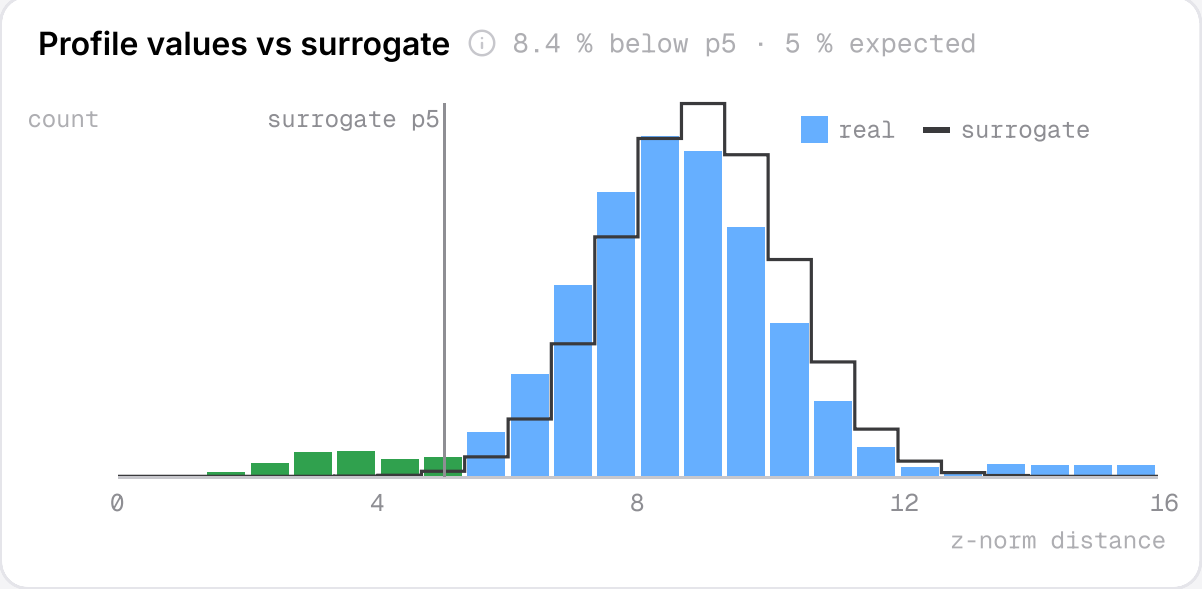
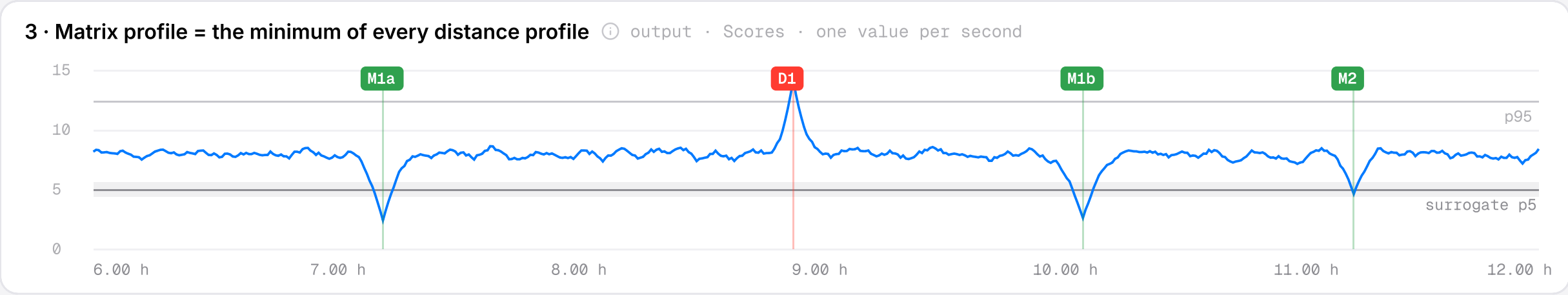
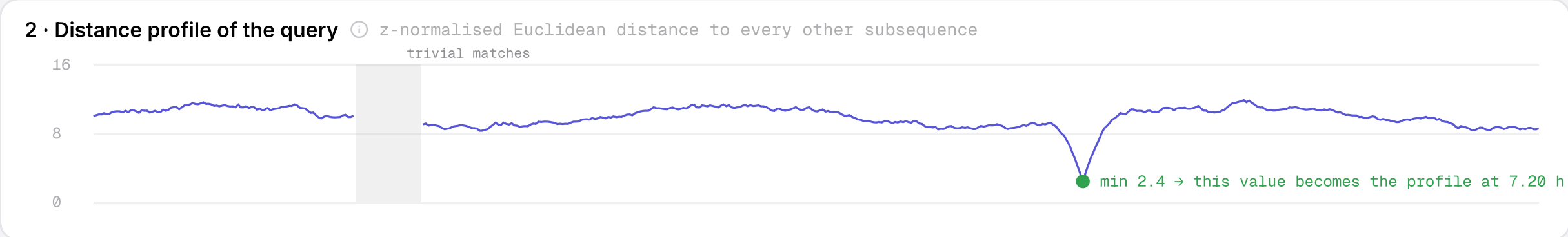
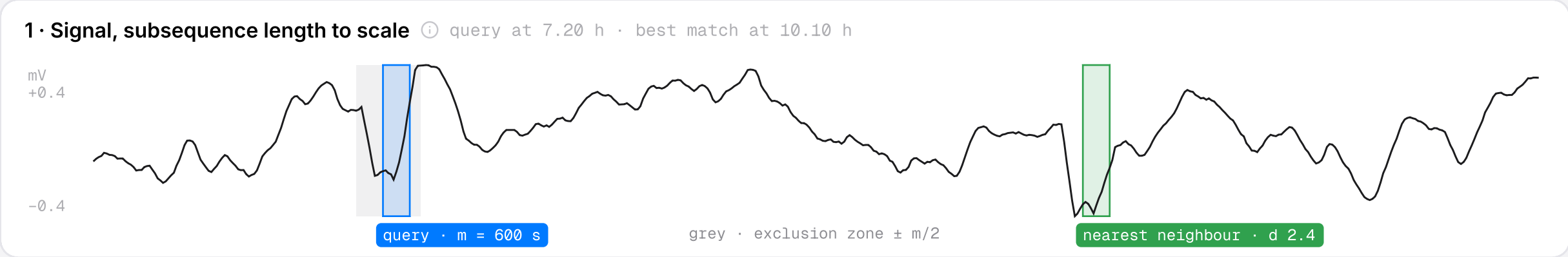
Save template

Re-run 01

chain

- Source
 - Signal

cached>01 Matrix profileSignal → Scorescached+



Parameters

changing any re-runs 01

subsequence length m 600 s

recommended 540 s · 3 × longest expected event

exclusion zone m / 2 top-k reported 3 motifs · 1 discord

normalisation z-norm raw algorithm STUMP · exact

surrogate IAAFT · 200× null reference p5 / p95

Top locations

click to centre panels 1-3

	at	distance	vs null
M1	7.20 h ↔ 10.10 h	2.4	p < 0.01
M2	10.83 h ↔ 6.52 h	4.6	p 0.04
M3	8.28 h ↔ 11.37 h	5.3	n.s. 0.21
D1	8.90 h	14.2	p < 0.01

this span ≈ 4 minon disk 0.2 MBwhole channel 6.4 h

whole channel exceeds the local ceiling → HPC

SLURM script+ Insert threshold →